

■ PolyGamma

`PolyGamma[z]` gives the digamma function $\psi(z)$.

`PolyGamma[n, z]` gives n^{th} derivative of the digamma function $\psi^{(n)}(z)$.

`PolyGamma[z]` is the logarithmic derivative of the gamma function, given by $\psi(z) = \frac{\Gamma'(z)}{\Gamma(z)}$. ■ `PolyGamma[n, z]` is given by $\psi^{(n)}(z) = \frac{d^n}{dz^n} \psi(z)$. ■ The digamma function is $\psi(z) = \psi^{(0)}(z)$: $\psi^{(n)}(z)$ is the $(n+1)^{\text{th}}$ logarithmic derivative of the gamma function. ■ See page 364. ■ See also: `Gamma`, `EulerGamma`.